# interdomain routing

so far: routing within one organization

lots of trust/sharing:

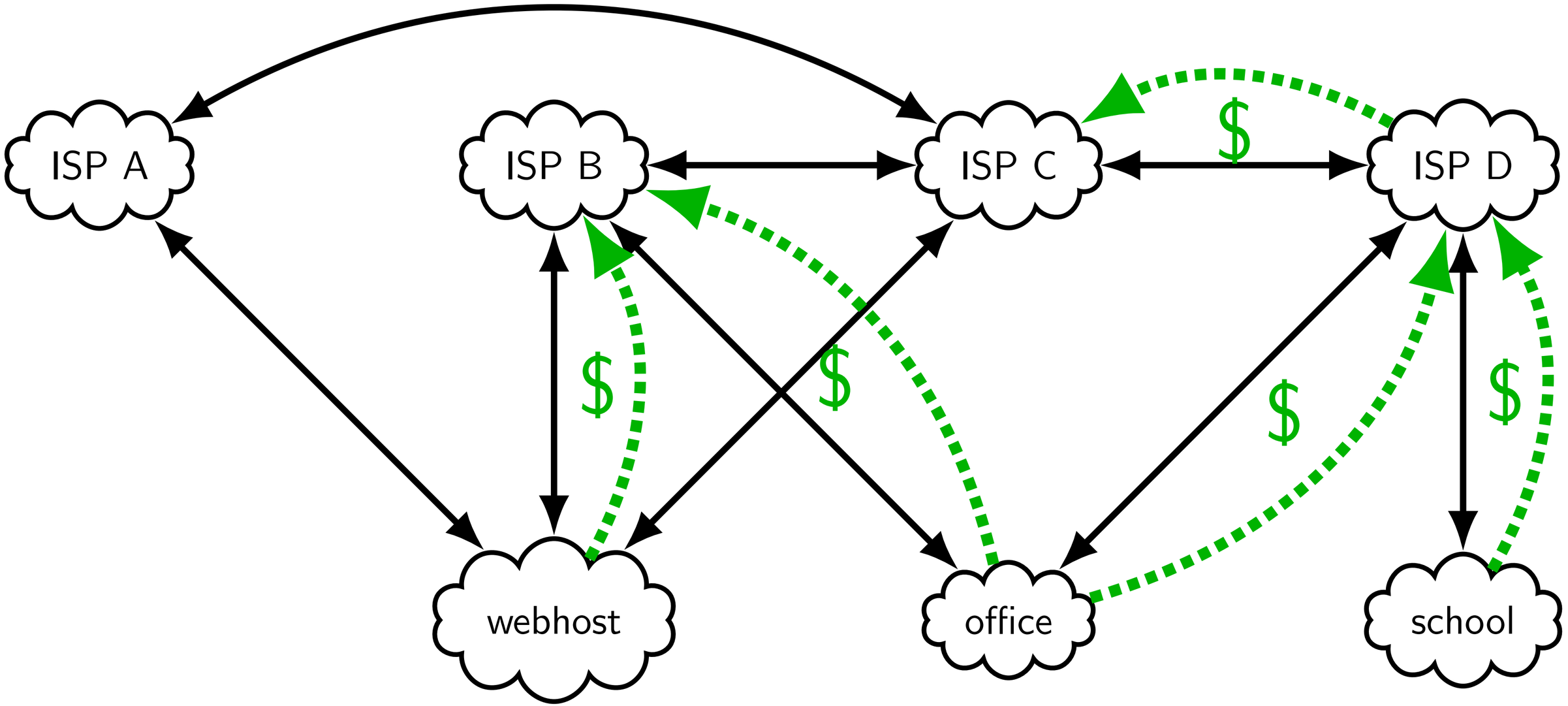
okay to send packets through (essentially) every router

okay for any router to ‘announce’ any address

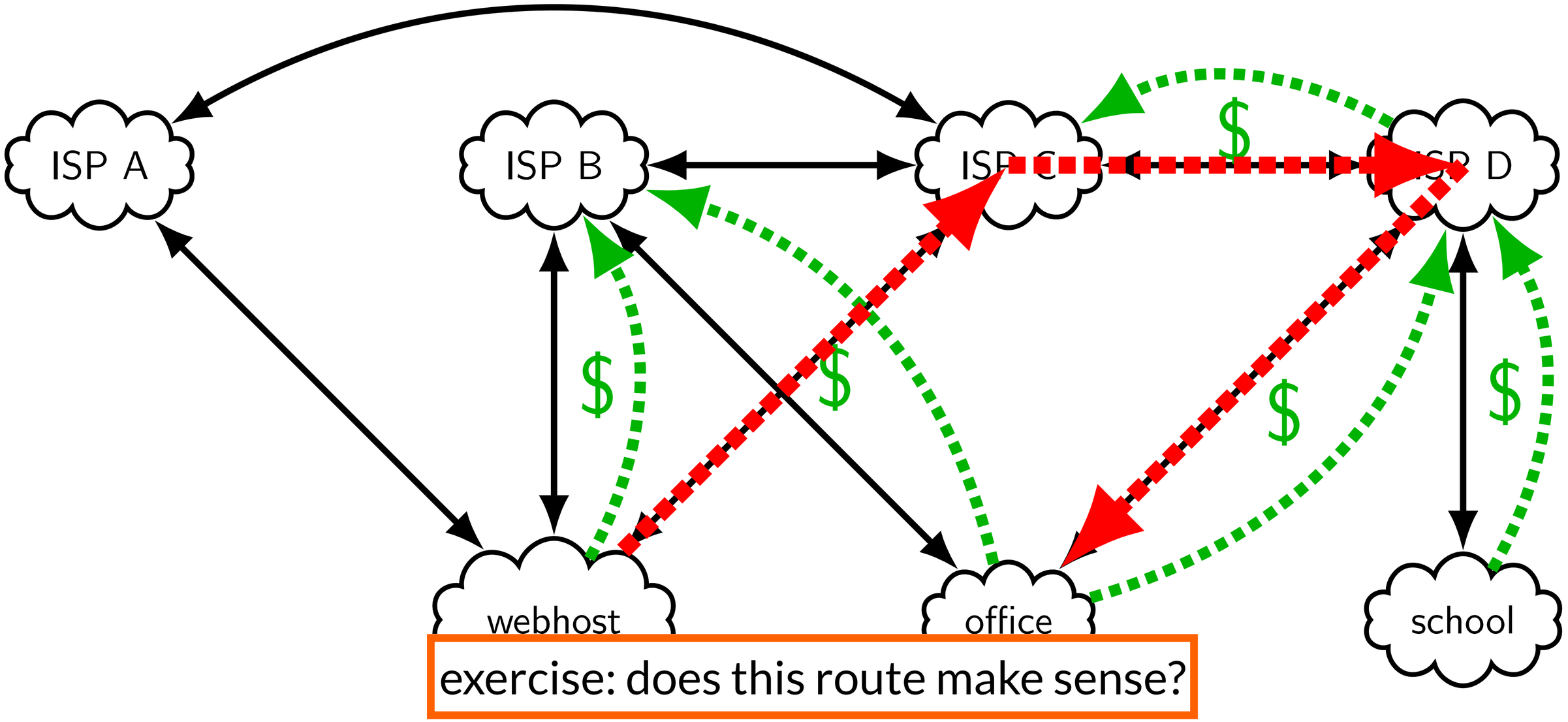
okay to share (almost) full map of network

not what we want for interdomain routing

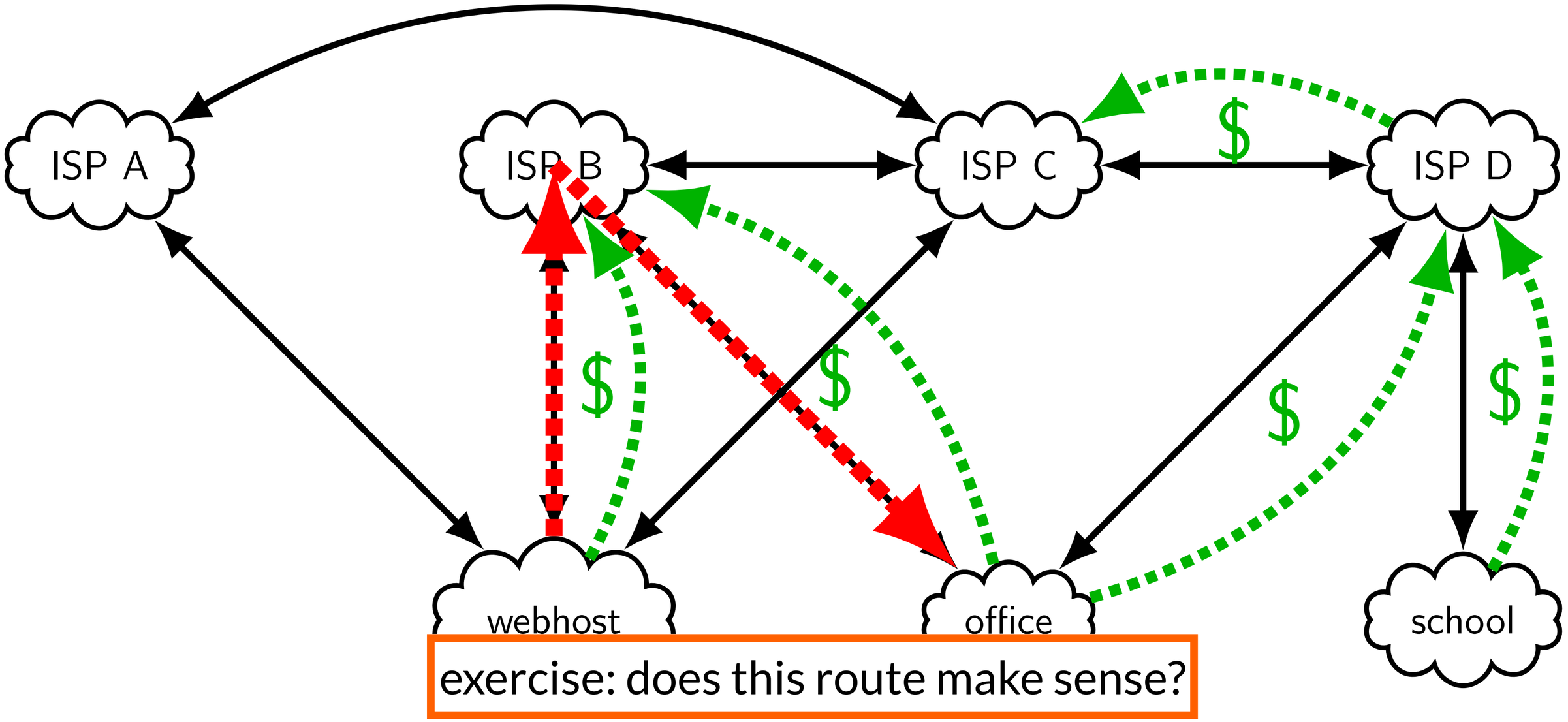
# some business considerations



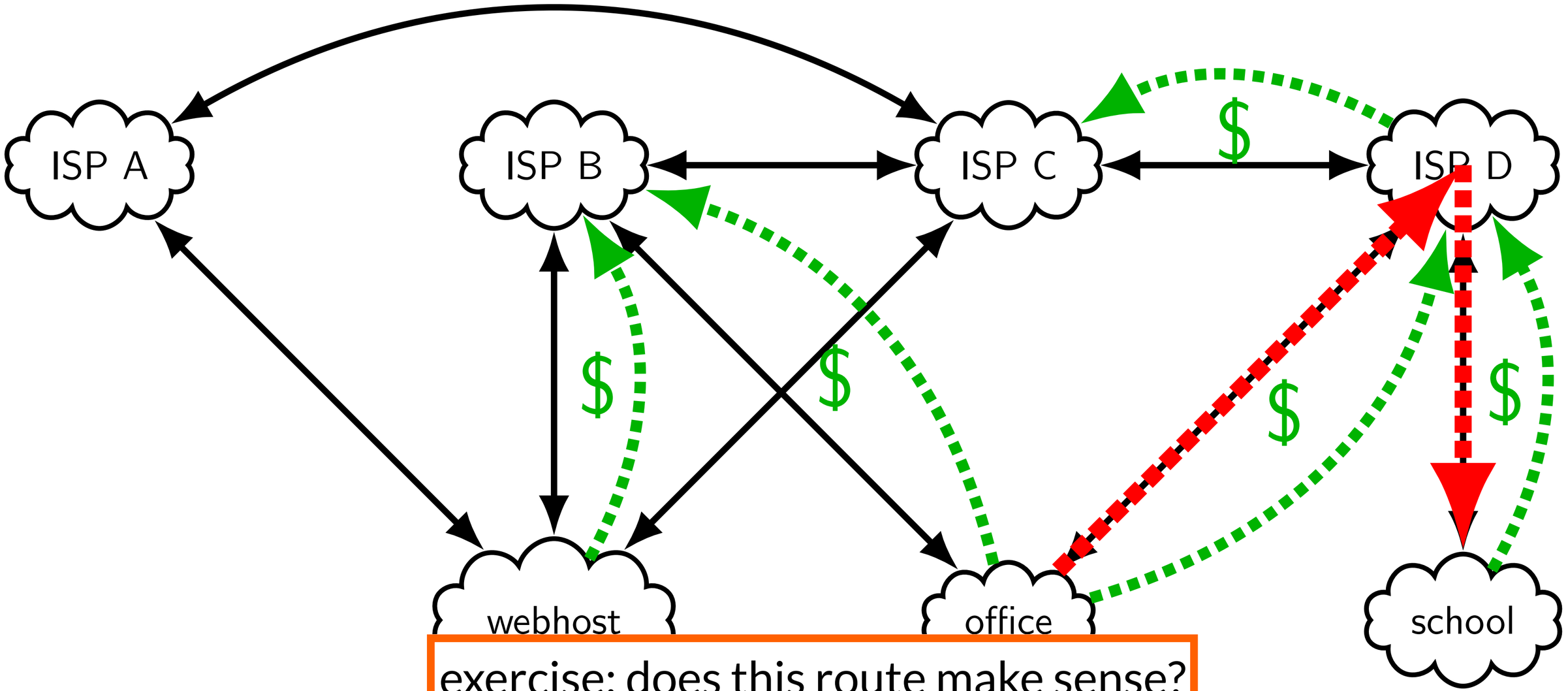
# some business considerations (exercise 1)



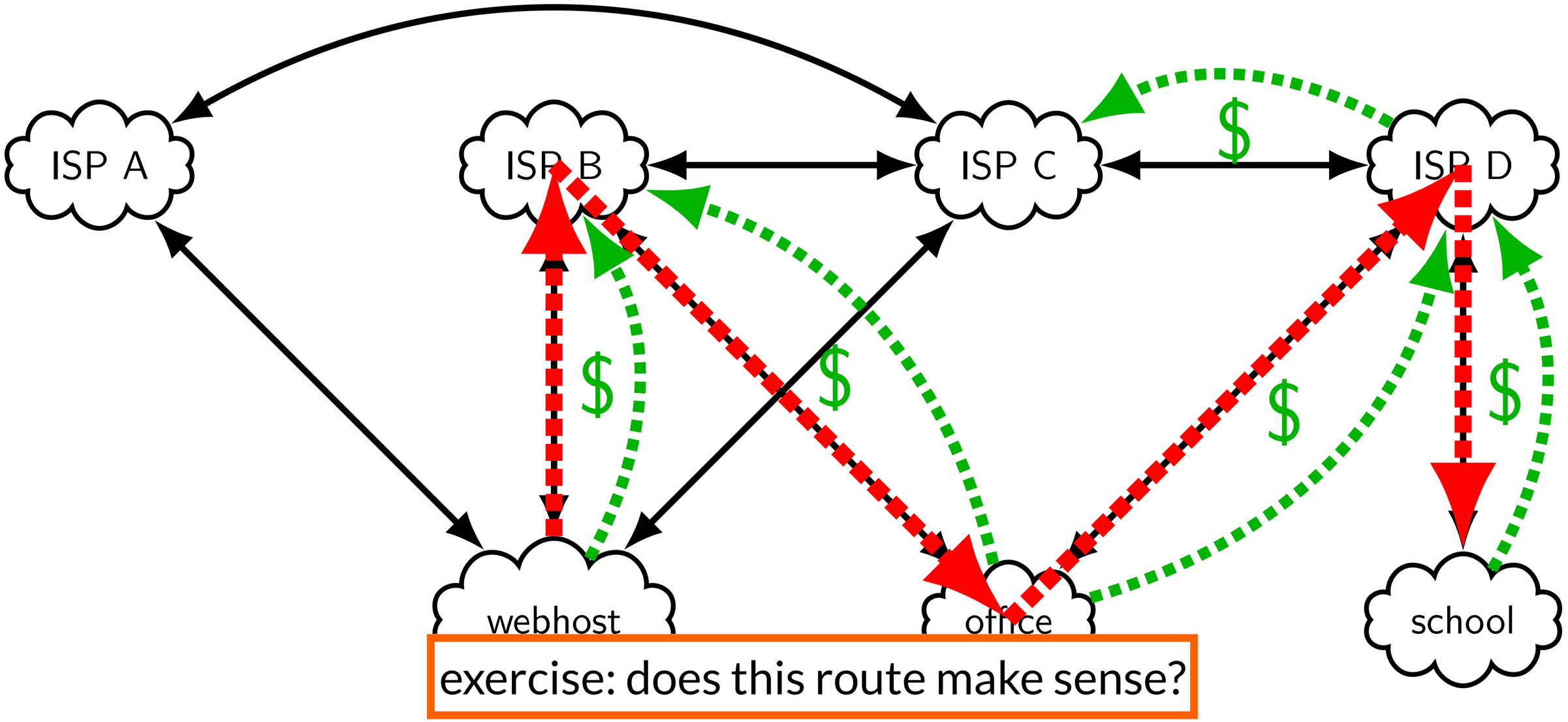
# some business considerations (exercise 2)



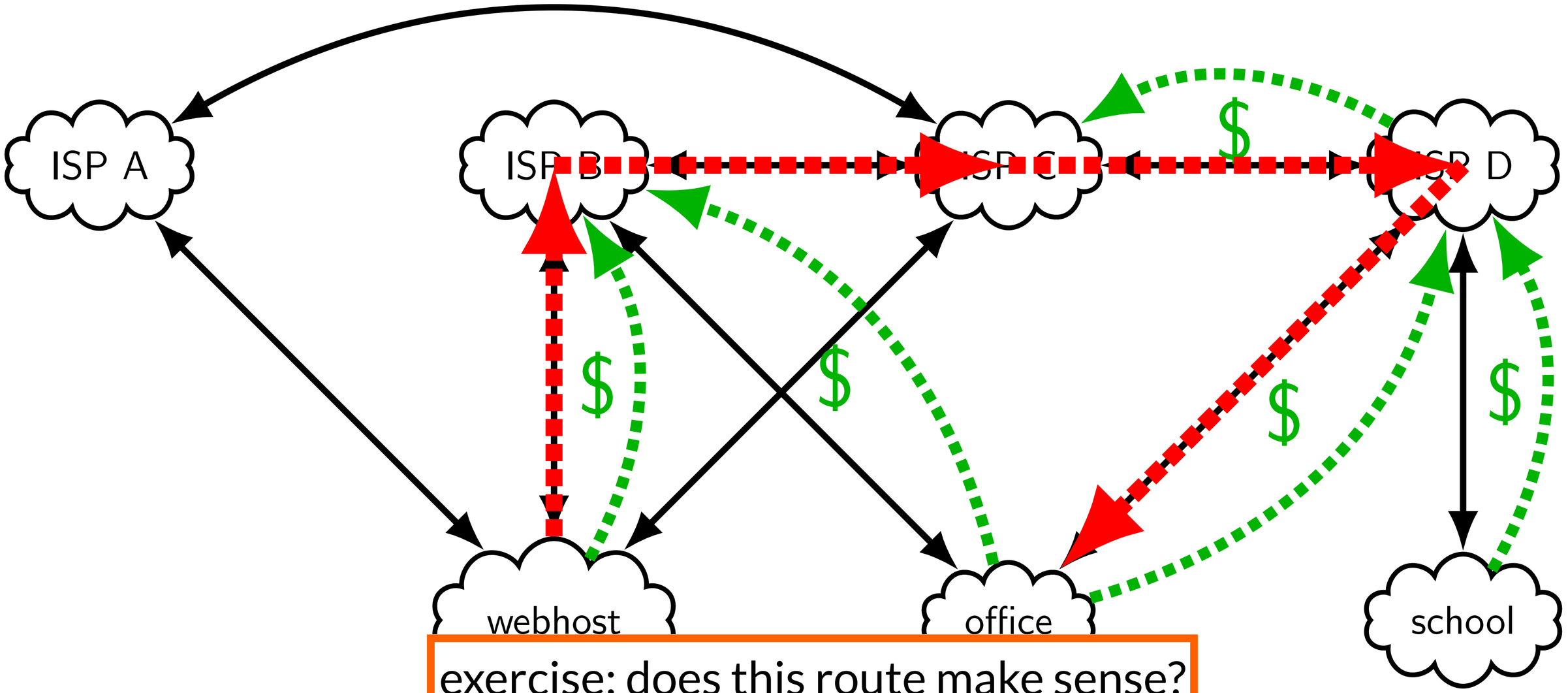
# some business considerations (exercise 3)



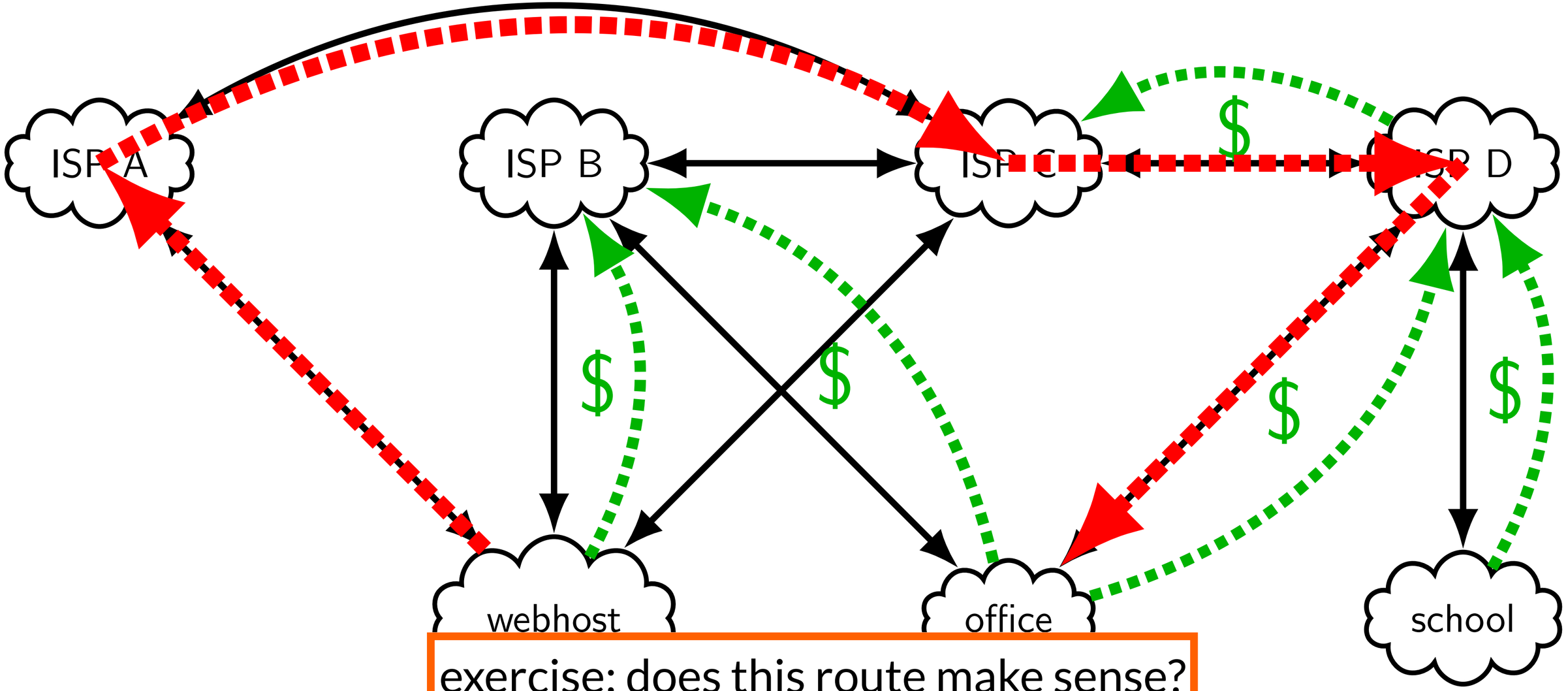
# some business considerations (exercise 4)



# some business considerations (exercise 5)



# some business considerations (exercise 6)



# autonomous system

autonomous system (AS) — one “routing domain”

- typically = set of networks administrated by one organization
- decides what routing to use internally
- should be fully connected internally

scope of OSPF instance = one AS

each AS can connect to other ASes

- well-defined protocol for sending routes to other ASes

# AS numbers

for Internet routing, ASes are assigned numbers

assigned by IANA and RIRs (similar to IP addresses)

originally 16-bit, now extended to 32-bit

some private use / special AS numbers

# relationship types

provider/customer

- typically: customer pays provider
- provider connects customers everywhere it can (customer paid for it)
- customer does *not* provide paths through its network

peer/peer

- often: no payment (‘settlement-free’)
- if A peers with B…
- A gets connected to B’s customers (customers paid B for this)
- A does not get connected to B’s other peers (no one paid B for this)
- A does not get connected to B’s providers (no one paid B for this)

# connecting big networks?

some options:

(which are basically the same as connecting parts of big network)

run a fiber between two buildings

permitting and construction needed

pay for direct access to fiber someone else ran (“dark fiber”)

burying one fiber costs similar to burying bundle, so spares

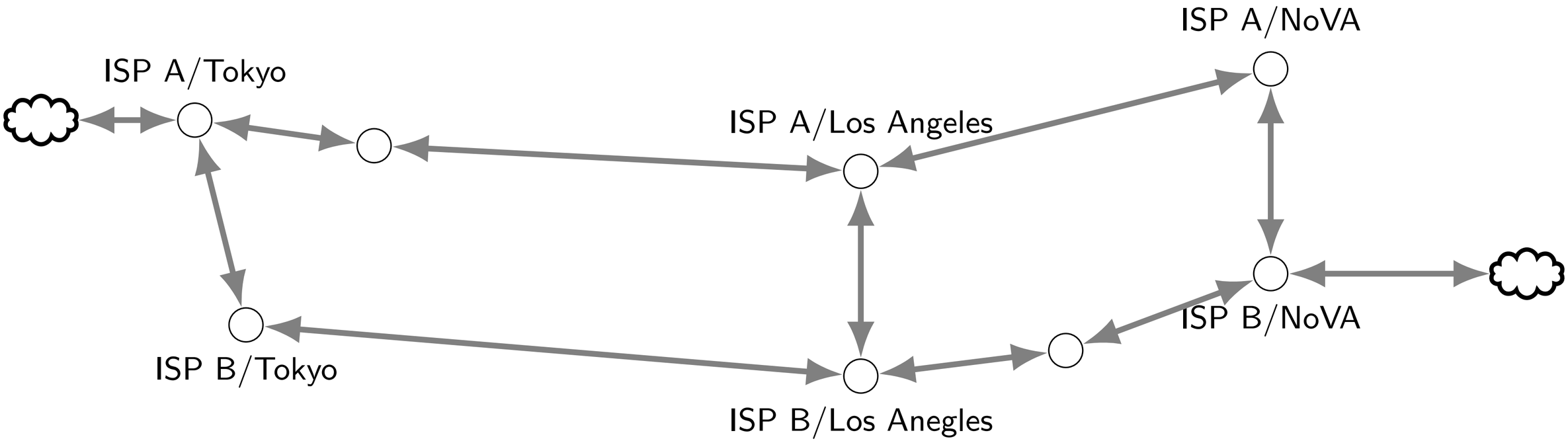
pay a telecom for a site-to-site connection

“gaurenteed” bandwidth+latency between two sites

may or may not use series of dedicated fibers

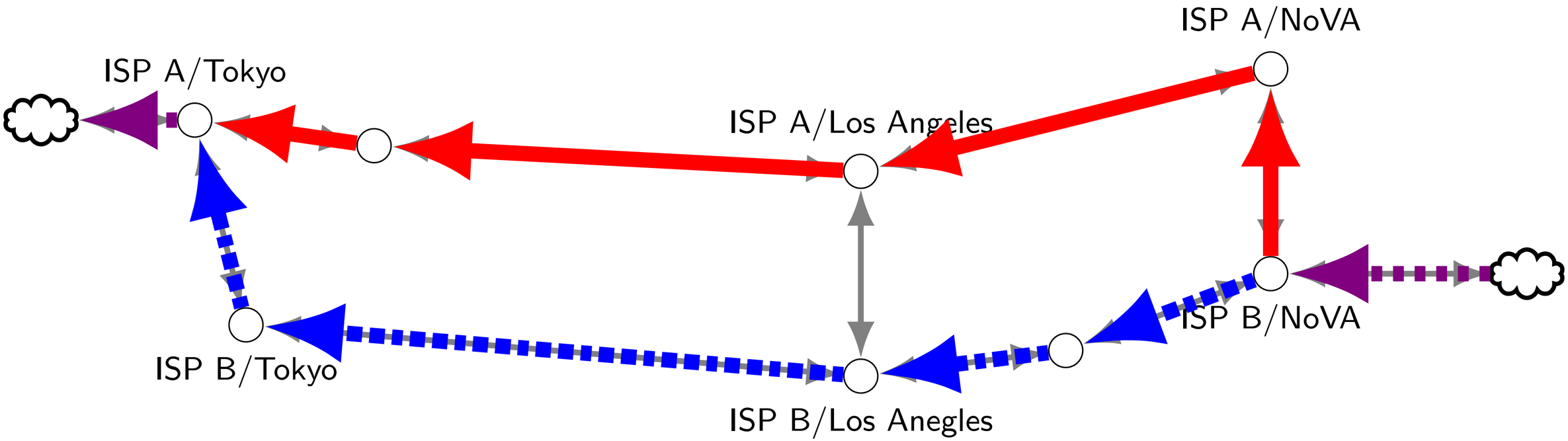
get space in common datacenter, pay datacenter operator for connection

# going the distance




does ISP A or ISP B help packets cross the Pacific?

# going the distance




does ISP A or ISP B help packets cross the Pacific?

# distance preferences

ISP B$\rightarrow$ISP A across the Pacific:

for ISP B:

- cheaper to hand-off packet to ISP A as soon as possible
- more control over performance if handing off as late as possible

for ISP A:

- cheaper to require ISP B to hand-off packet as late as possible
- more control over performance if B sends as soon as possible

maybe part of ISP A and ISP B peering agreement

# Border Gateway Protocol

protocol for sending routes between networks

used whereever routers from different ASes connect

“EBGP”

used within AS to share routes out of AS internally

“IBGP”

each router constructs list of routes to offer

each router receives list of routes, exports to OSPF/etc.

# BGP connections

BGP (TCP) connections made between routers

each router keeps track of set routes advertised by other

command sent to add or withdraw specific routes

not like distance vector where we kept resending everything

# BGP prefixes

routes sent via BGP called ‘prefixes’

said to be “announced” from one router to another

because the network (e.g. 10.0.1.0/24) is the important part

(and the next hop is implied by where prefix comes from)

# BGP route

adjacent routers share list of *routes* with:

IP prefix (CIDR-style, basically)

AS path — list of autonomous system the route goes through

next hop router (IP address)

*multi-exit discriminator*

low value = this entrance to AS is better than others for these IPs

*local preference* (internal-only)

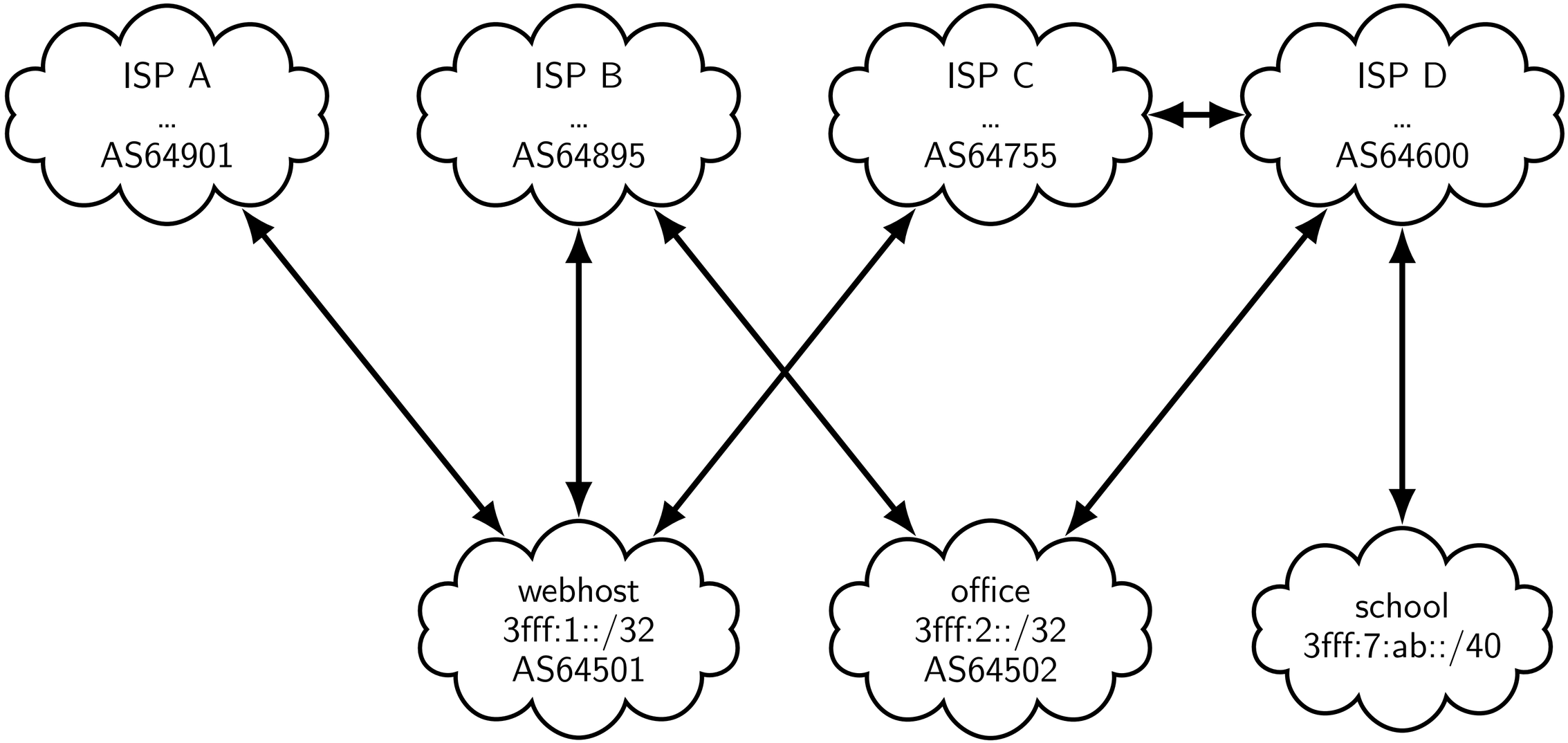
# AS path

used to detect routing loops

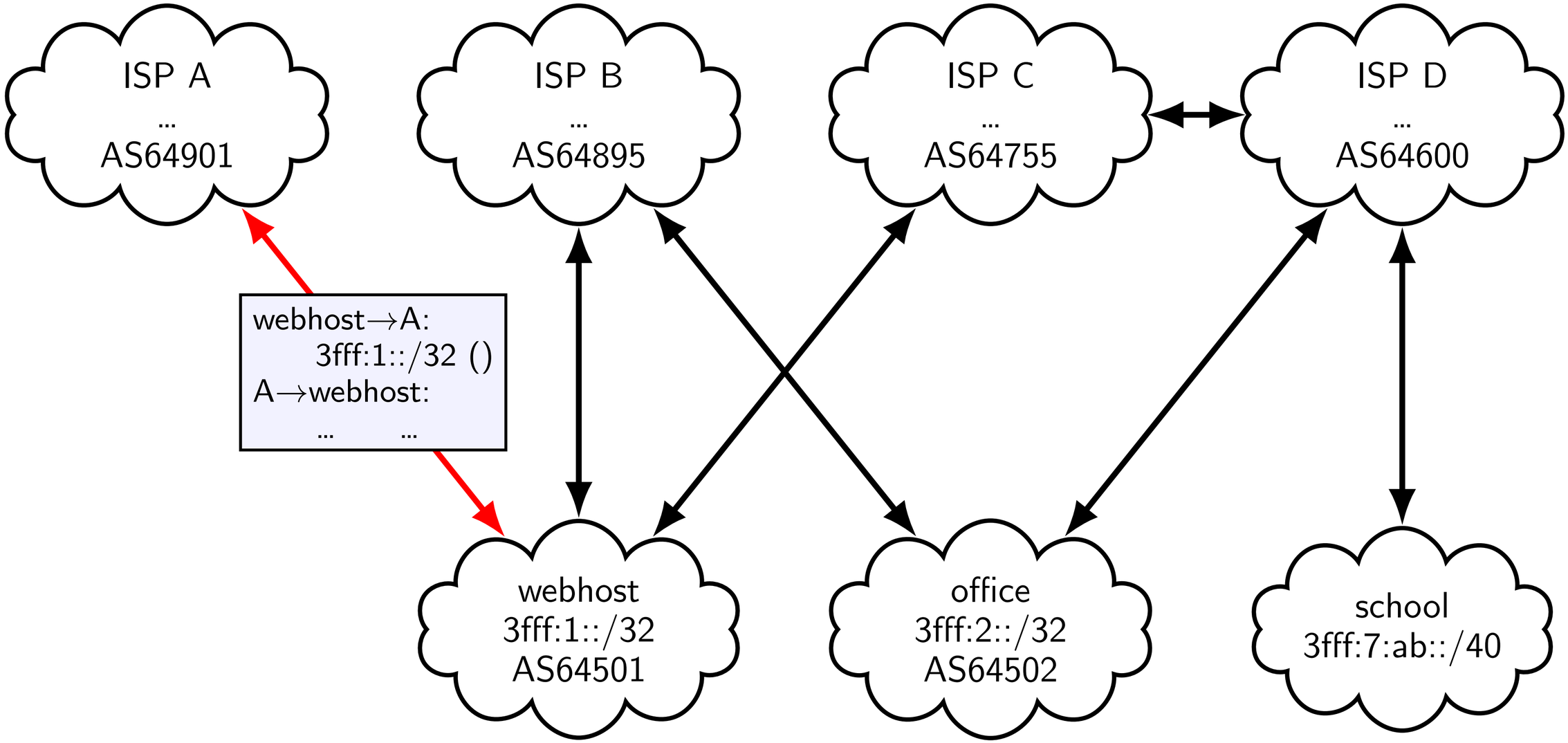
append your AS when sending route externally

always ignore external routes with your AS in their AS path already

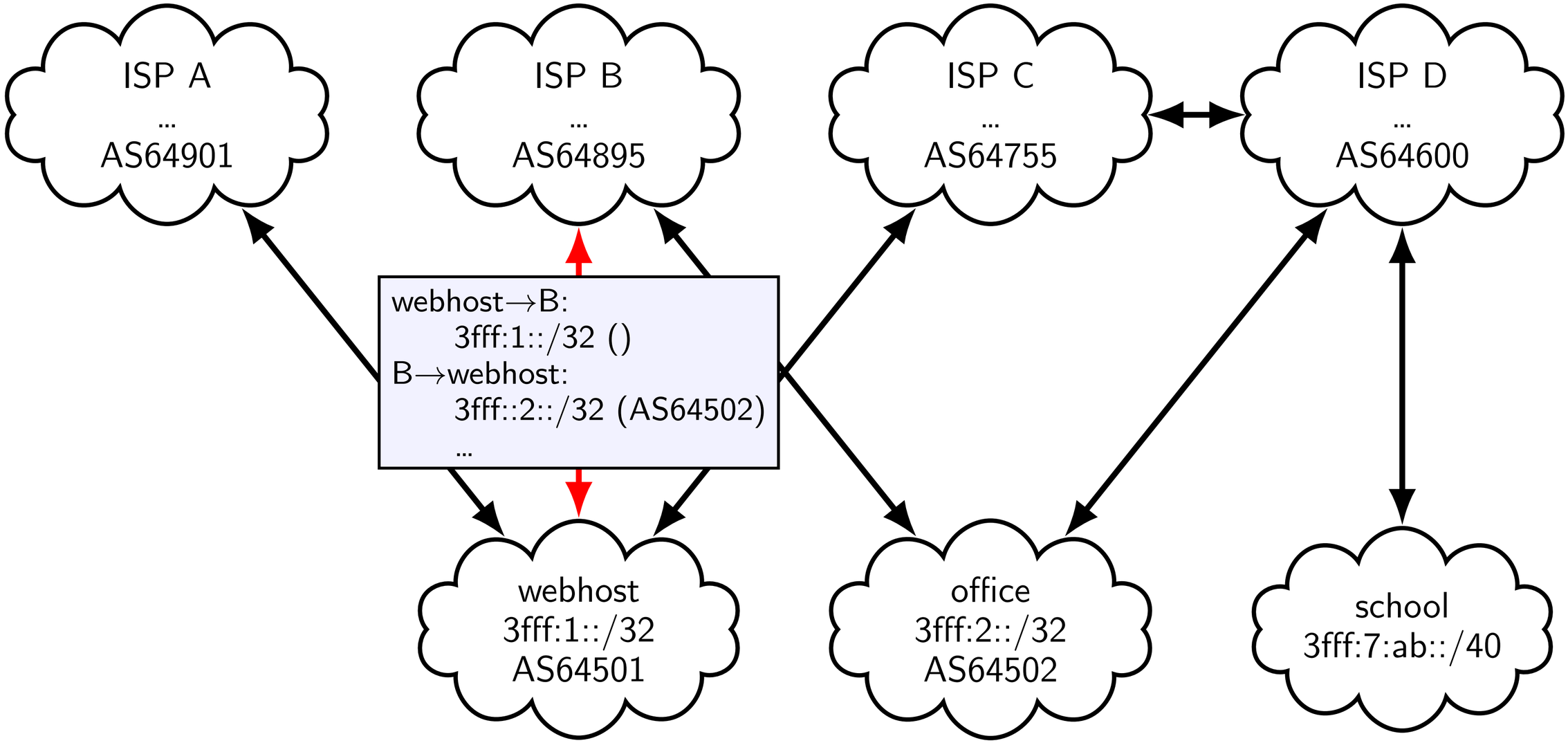
# external BGP



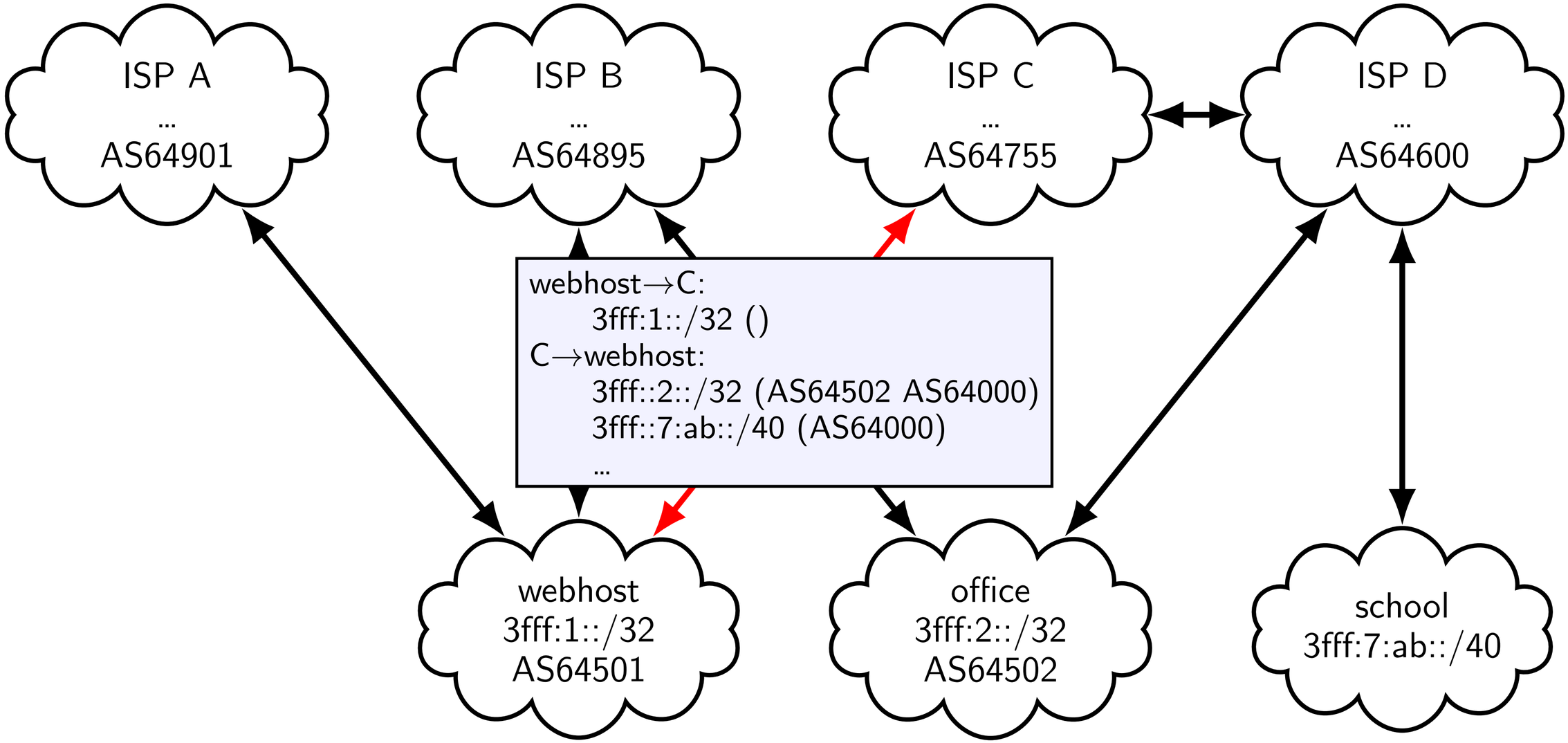
# external BGP



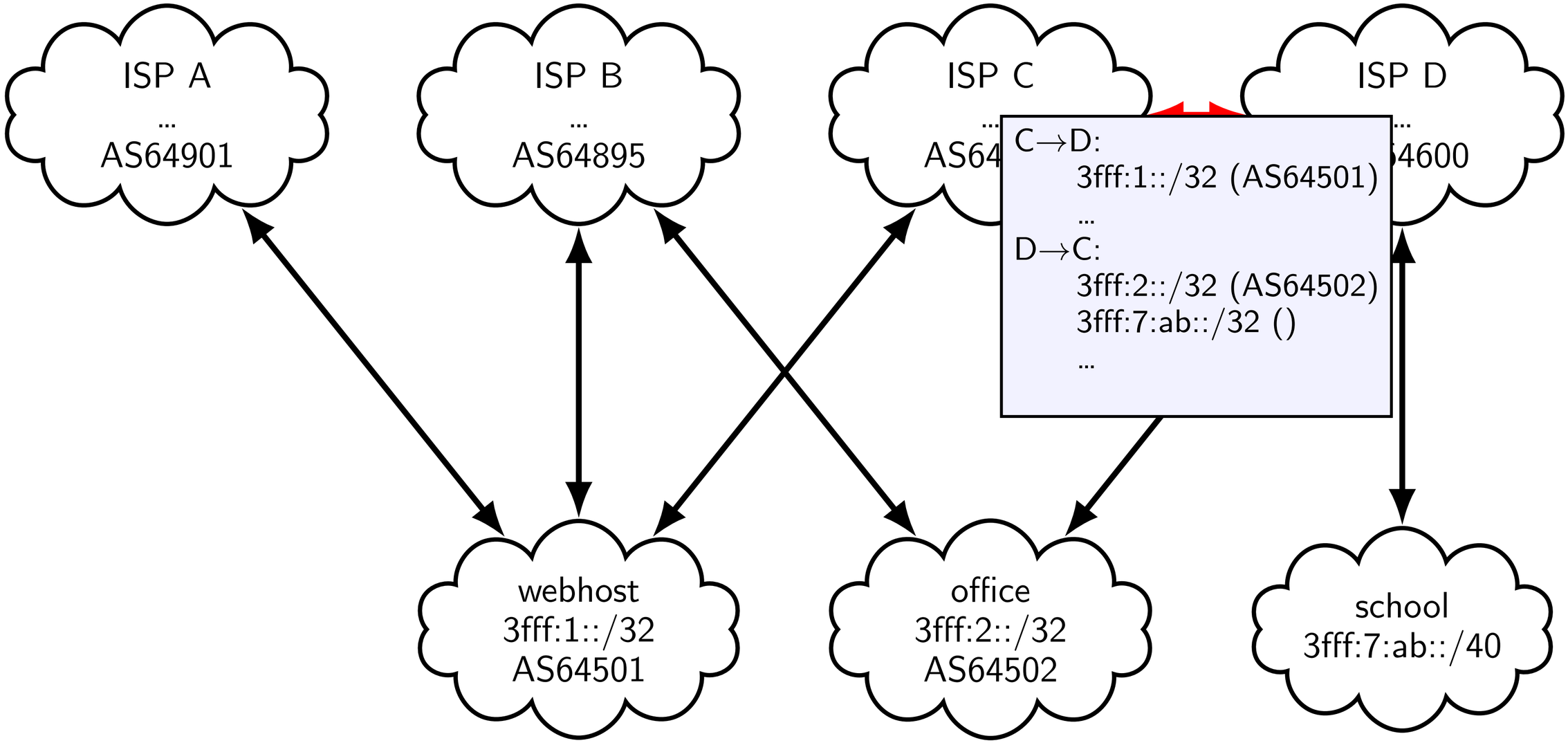
# external BGP



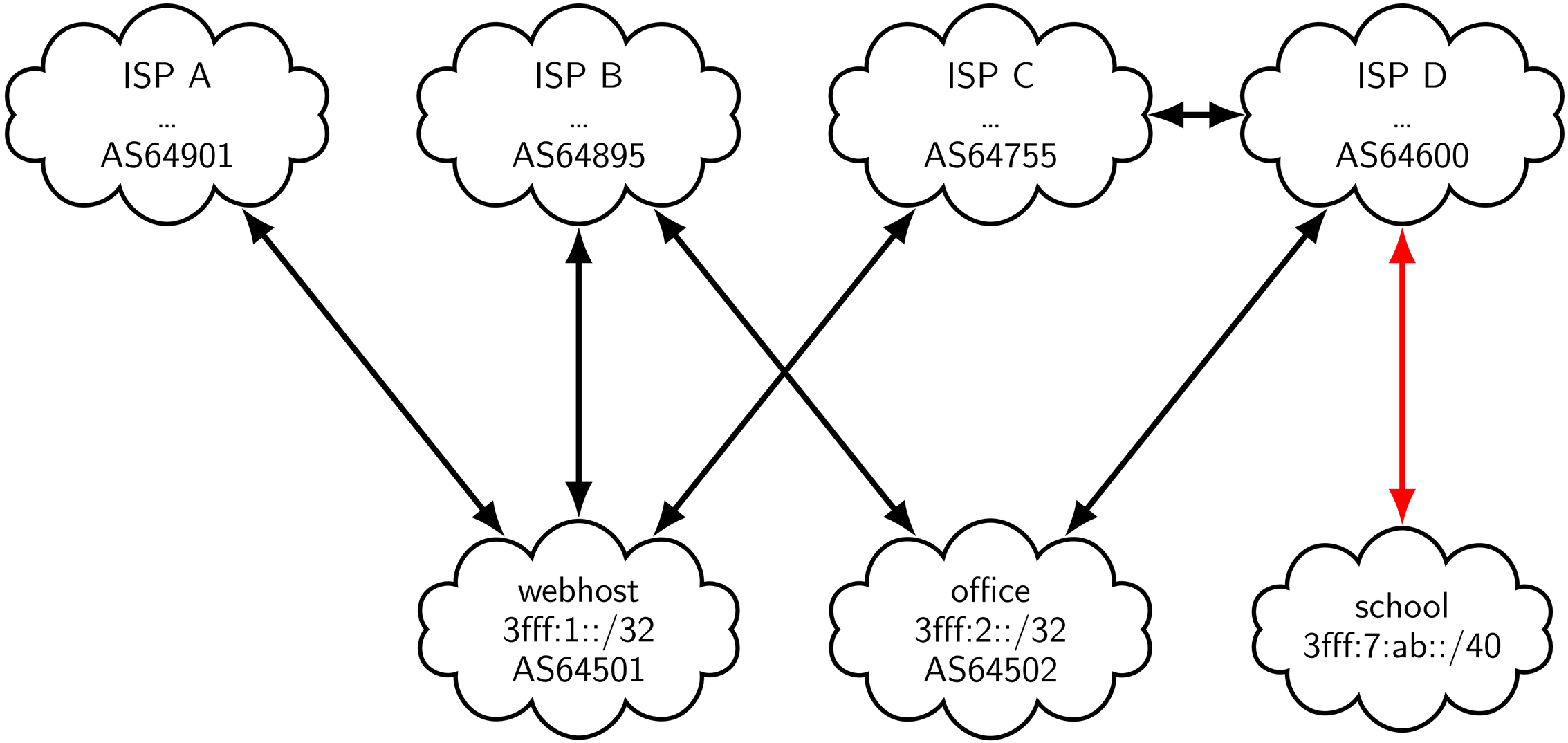
# external BGP



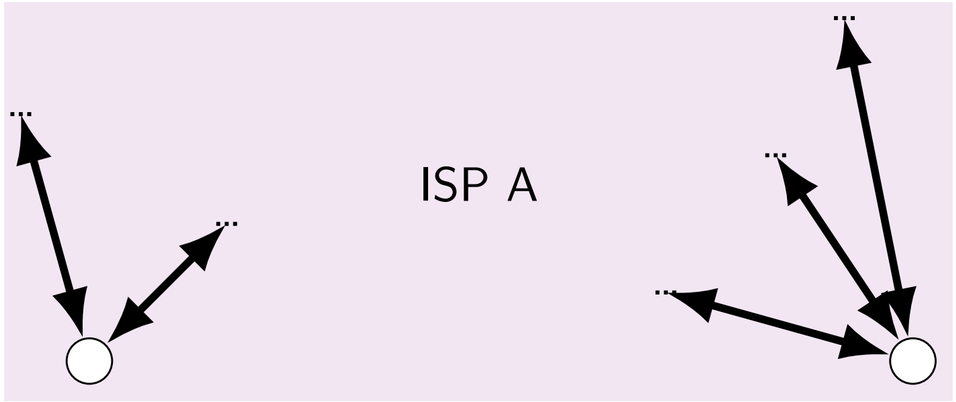
# external BGP

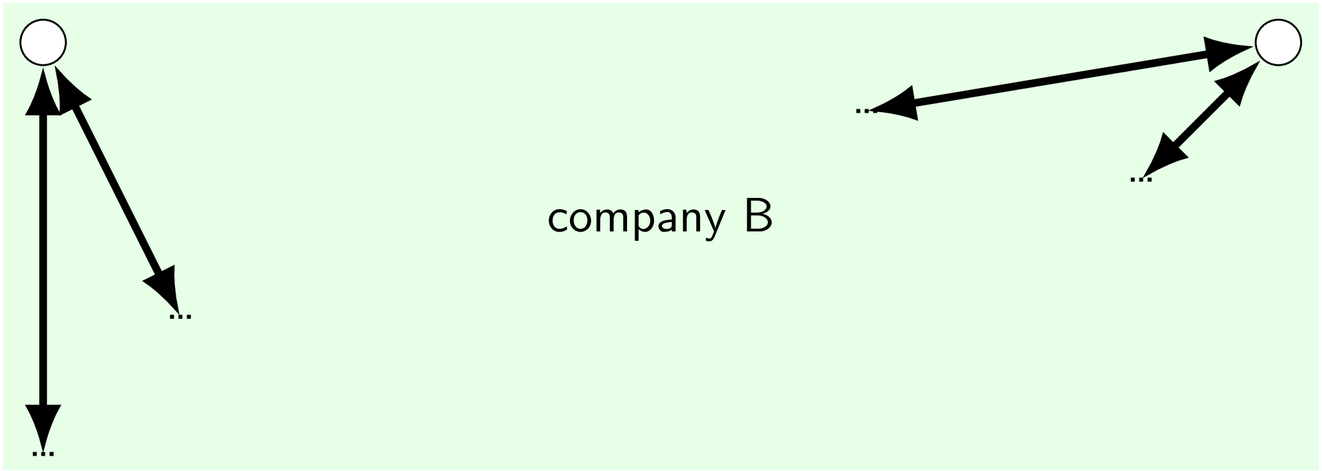


# external BGP

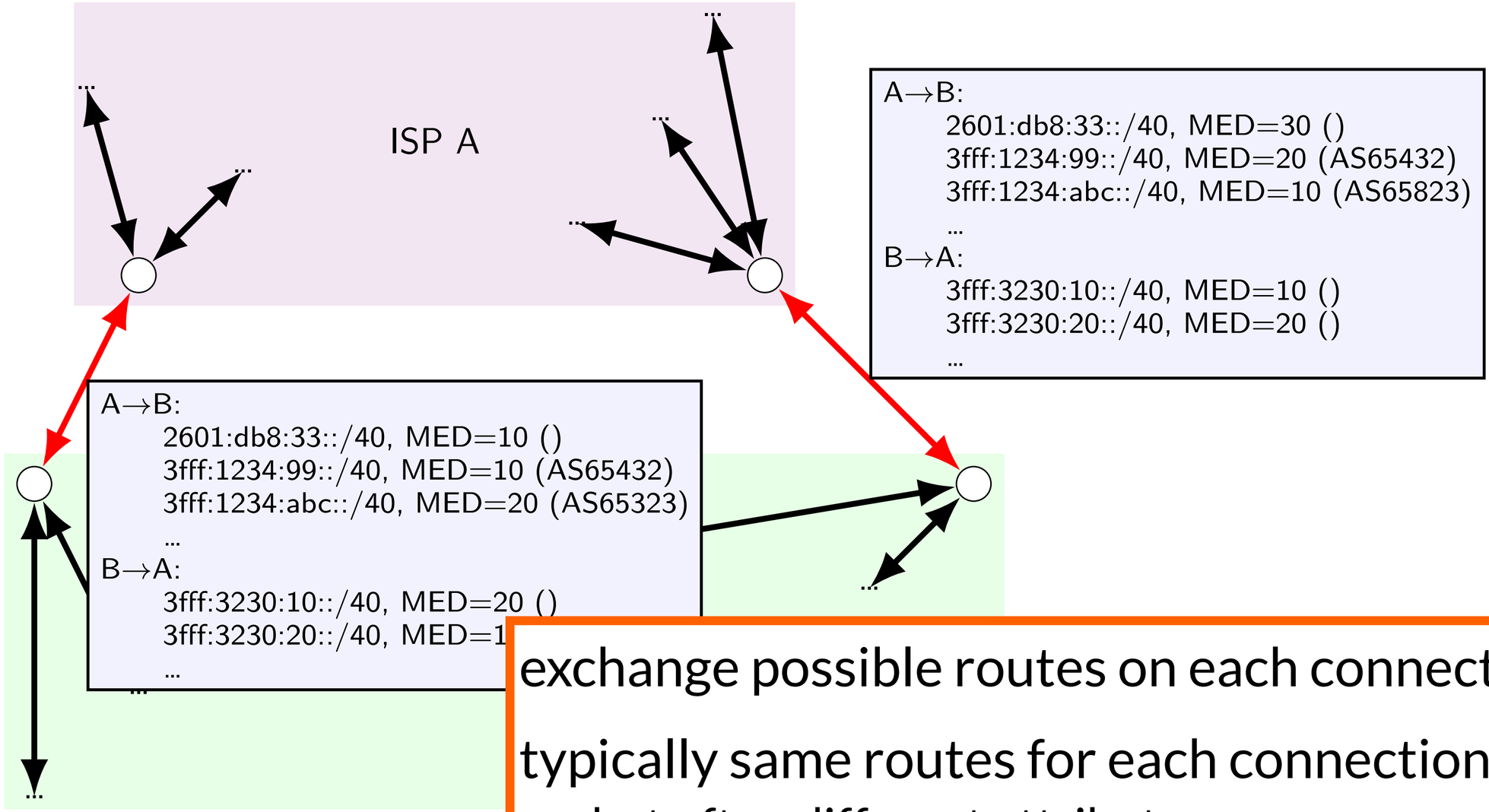


# multiple BGP sessions



# multiple BGP sessions



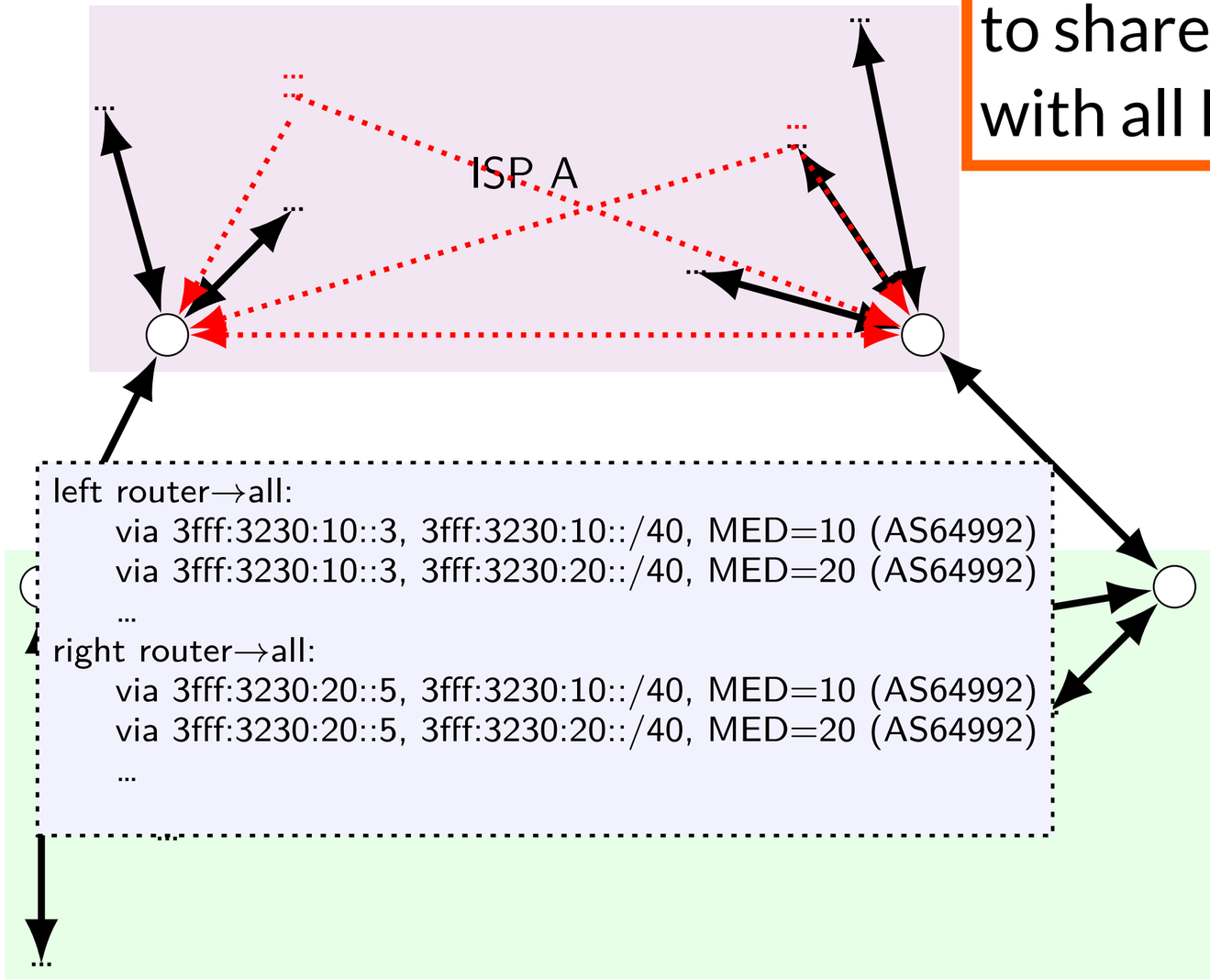

exchange possible routes on each connection

typically same routes for each connection to AS

but often different attributes

# multiple BGP sessions

within ISP, use internal BGP (IBGP) to share everytig learned via BGP with all BGP routers



# preference between routes

if multiple choices, most common strategy:…

should use most specific route

use 2001:db8:1234::/40 over 2001:db8:1234::/39 if both apply

(but usually reject very small address ranges (e.g. /31 for IPv4, /60 for IPv6))

then (if tie) local policy applies

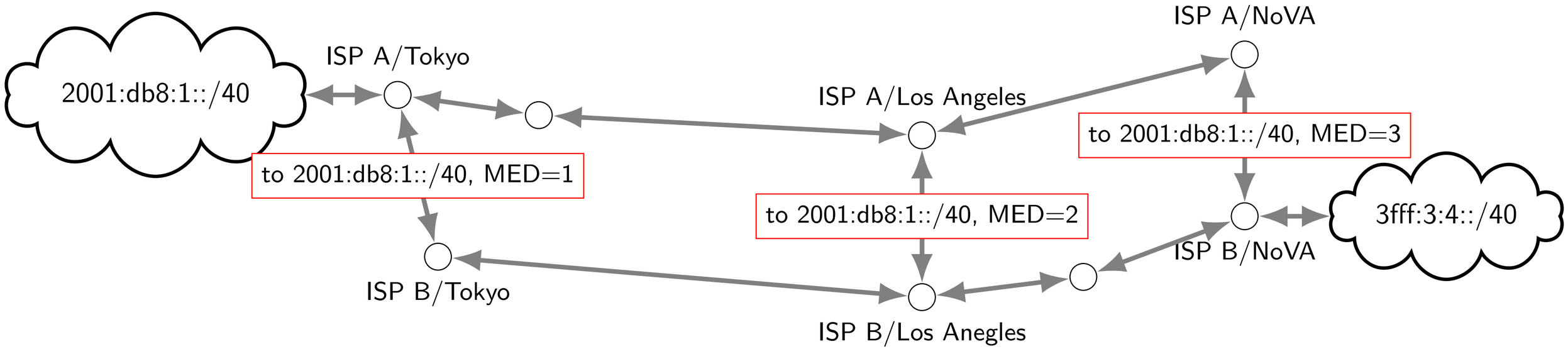
then shortest AS path

then lower AS number

then (sometimes) lower MED (multiple exit discriminator)

then best route within current AS

# multi-exit discriminator



# getting your preference

to affect how people route you, can…

prepend to AS path sent to make it longer

- typically add serveral copies of your AS number

only announce network from certain of your routers

- problem: won't have all 'backup' paths available

announce a large network in more specific pieces

- 3fff:1234::/32 as 3fff:1234::/33 and 3fff:1234:8000::/33

get other networks to change how they forward your routes

- often enabled through 'BGP communities'

https://bgp.he.net/super-lg/

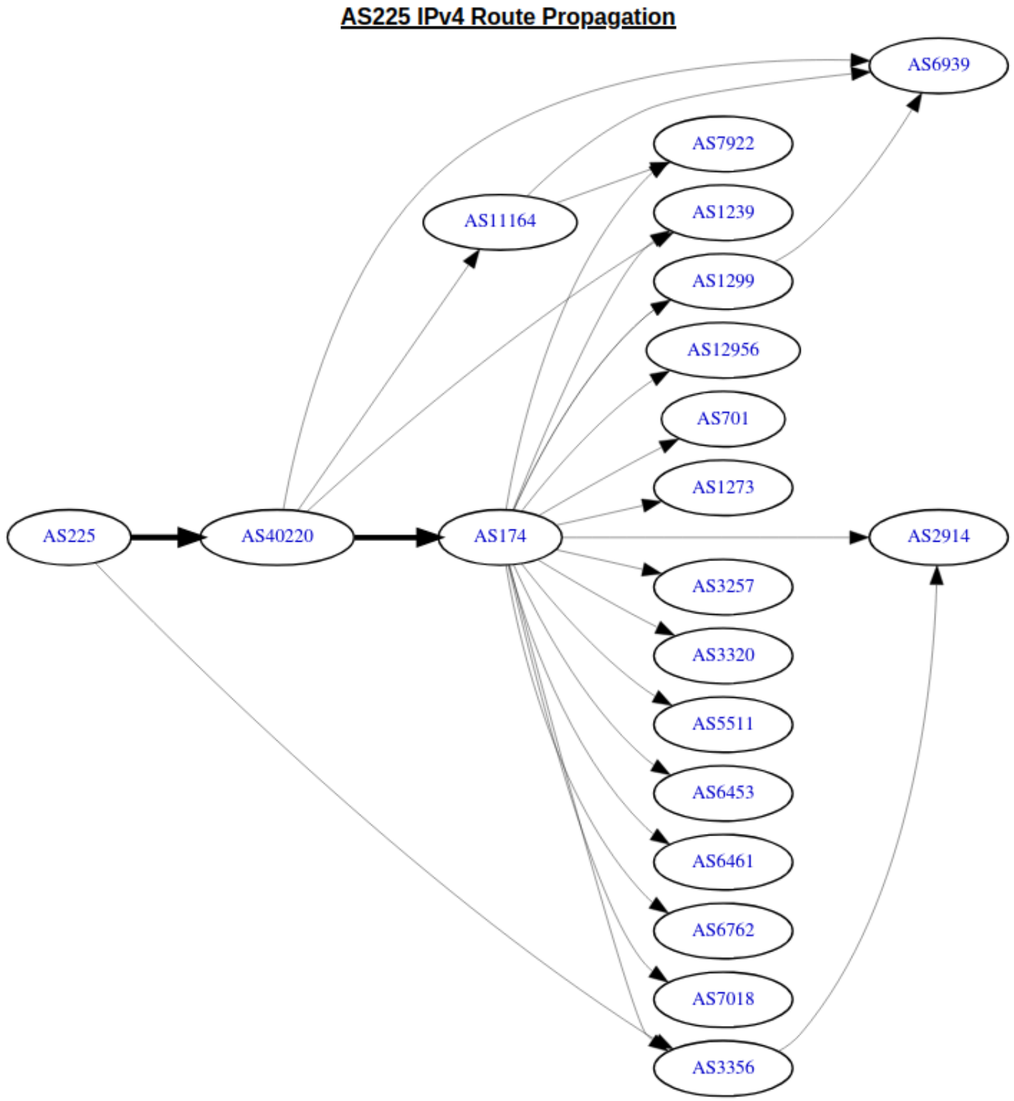
https://bgp.he.net/super-lg/#128.143.0.0/16?tob=none&mt=include&ma=6939&els=exact

https://lg.ring.nlnog.net/prefix?q=128.143.0.0/16&match=exact&peer=all

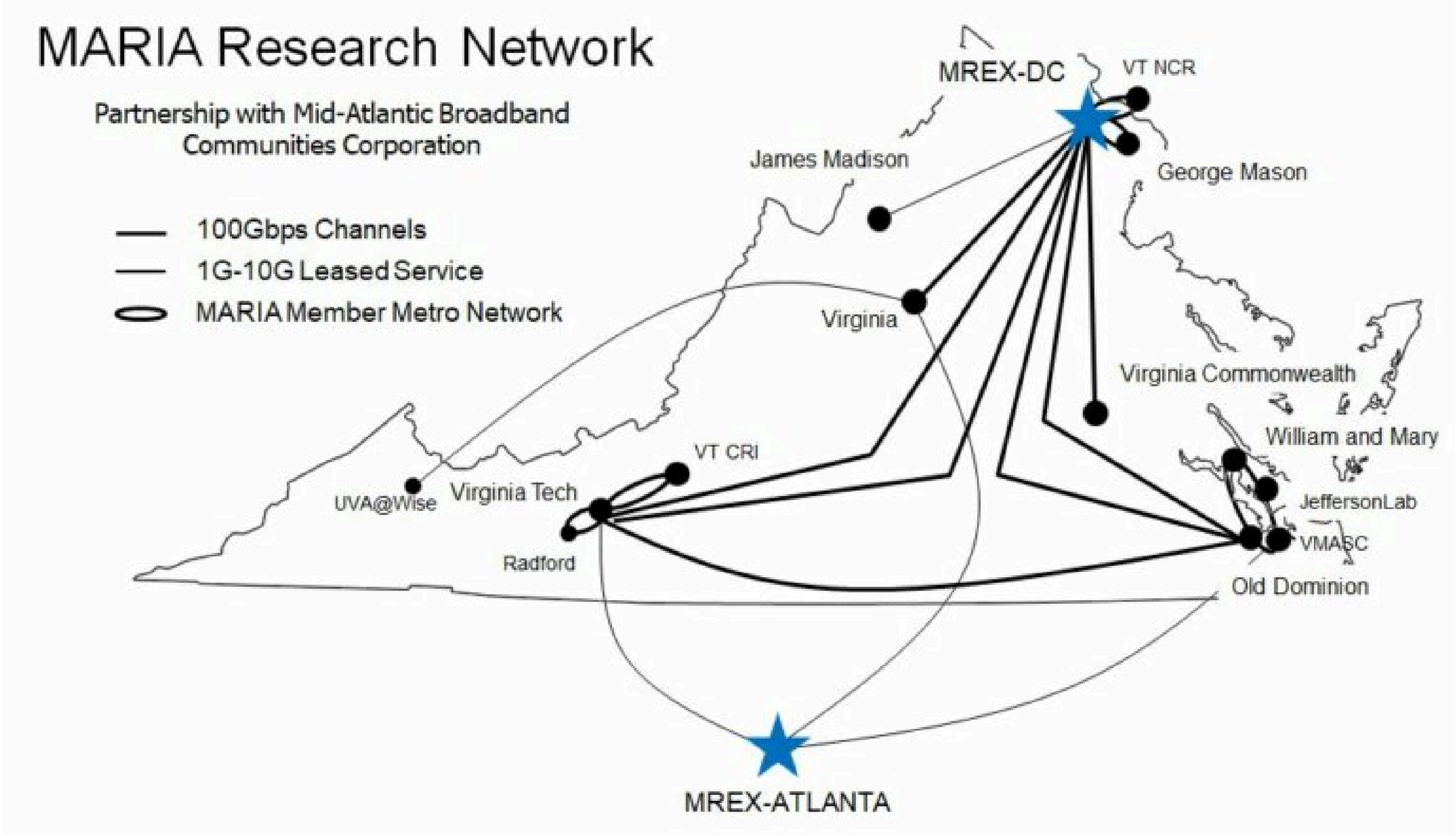
future slides mostly from 2024

UVa has changed their network configuration since then

# https://bgp.he.net/AS225 (University of Virginia)



# AS40220


MARIA Research Network
Partnership with Mid-Atlantic Broadband Communities Corporation
100Gbps Channels
1G-10G Leased Service
MARIA Member Metro Network
MREX-DC
VT NCR
James Madison
George Mason
Virginia
Virginia Commonwealth
William and Mary
VT CRI
UVA@Wise
Virginia Tech
JeffersonLab
VMASC
Radford
Old Dominion
MREX-ATLANTA

PeeringDB

Search here for a network, IX, or facility.

Advanced Search Legacy Search

Register or Login

English (English)

# Mid-Atlantic Terascale Partnership - MATP

EXPORT

| | |
|---|---|
| Organization | Mid-Atlantic Terascale Partnership - MATP |
| Also Known As | MARIA / Virginia Tech (Virginia Polytechnic Institute and State University) |
| Long Name | |
| Company Website | |
| ASN | 40220 |
| IRR as-set/route-set | AS40220:AS-ALL |
| Route Server URL | |
| Looking Glass URL | |
| Network Types | Educational/Research |
| IPv4 Prefixes | 90 |
| IPv6 Prefixes | 25 |
| Traffic Levels | Not Disclosed |
| Traffic Ratios | Mostly Inbound |
| Geographic Scope | Regional |
| Protocols Supported | ✓ Unicast IPv4 ○ Multicast ✓ IPv6 ○ Never via route servers |
| Last Updated | 2022-07-27T05:34:22Z |
| Public Peering Info Updated | 2024-07-24T16:31:34Z |
| Peering Facility Info Updated | 2016-03-14T20:59:43Z |
| Contact Info Updated | 2020-01-22T04:24:10Z |
| Notes | MATP (AS40220) is a consortium of research institutions in Virginia, Maryland, and Washington formed to to support research activities that require next-generation high-performance network connectivity. We have routers in Equinix Ashburn and TelX Atlanta and provide commodity and R&E network services to most research universities in Virginia. MATP is managed by Virginia Tech (AS1312). |

## Public Peering Exchange Points

Filter

| Exchange AŻ<br>IPv4 | ASN<br>IPv6 | Speed<br>Port Location | RS Peer | BFD Support |
|---|---|---|---|---|
| Digital Realty Atlanta<br>206.126.110.182 | 40220<br>2001:504:17:110::182 | 10G | ○ | ○ |
| Equinix Ashburn<br>206.126.236.139 | 40220<br>2001:504:0:2:0:4:220:1 | 20G | ○ | ○ |

## Interconnection Facilities

Filter

| Facility AŻ<br>ASN | Country<br>City |
|---|---|
| Digital Realty ATL (56 Marietta)<br>40220 | United States of America<br>Atlanta |
| Equinix DC1-DC15, DC21 - Ashburn<br>40220 | United States of America<br>Ashburn |

# AS3356

LUMEN®

SOLUTIONS PRODUCTS SHOP RESOURCES

DEDICATED INTERNET ACCESS

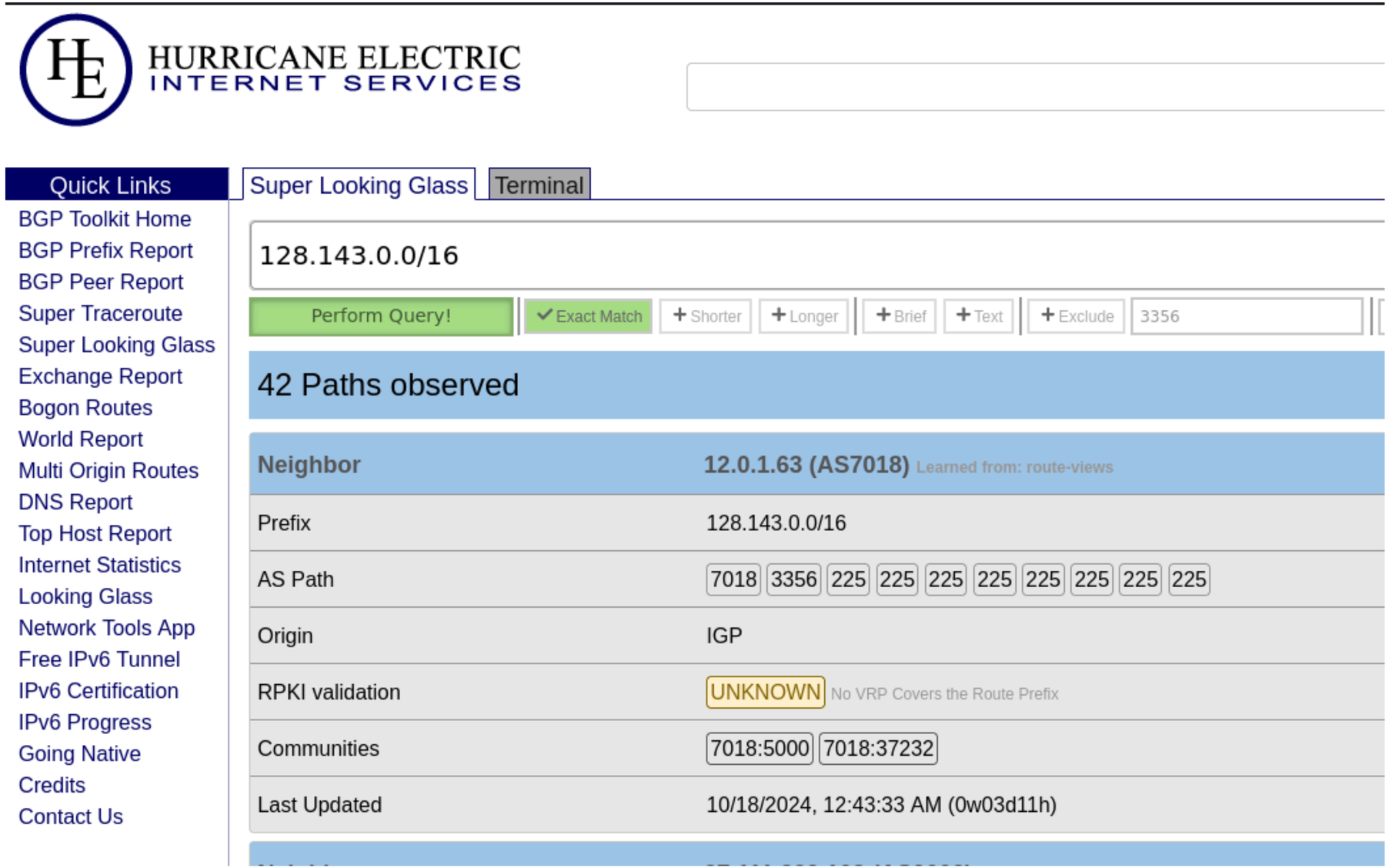
## Go full speed

Give your enterprise a direct fast lane to 6,300+ interconnects[1] with speeds up to 100 Gbps.

GET OFFER VIEW SPECS

Overview Benefits Pricing Customer Stories Resources

# 2024: AS3356 is a backup (8x prepend)



# 2025: AS3356 and AS 40220 equal?

| Neighbor | 172.18.14.1 (AS6939) Learned from: collector14.bgp.he.net |
|---|---|
| Prefix | 128.143.0.0/16 |
| AS Path | 6939 3356 225 |
| Origin | IGP |
| RPKI validation | UNKNOWN No VRP Covers the Route Prefix |
| Last Updated | 10/21/2025, 5:26:12 AM (0w00d08h) |

| Neighbor | 172.18.14.2 (AS6939) Learned from: collector14.bgp.he.net |
|---|---|
| Prefix | 128.143.0.0/16 |
| AS Path | 6939 40220 225 |
| Origin | IGP |
| RPKI validation | UNKNOWN No VRP Covers the Route Prefix |
| Last Updated | 9/30/2025, 10:19:43 AM (3w00d03h) |

# peeringdb

https://peeringdb.com — commonly used database of ASes and how to peer with them

there is also – “whois” records (from RIRs) for ASes, IP blocks with contact info

# internet exchanges and route servers

internet exchange

- local network (typically within metro area) for connecting networks
- often run at and/or by ‘carrier-neutral’ datacenter
- typically high bandwidth (10-100Gbps ports to network)

route servers

- BGP servers run by internet exchange
- consolidates routes from participants
- takes advantage of “next hop” not needing to be router advertisement (even for EBGP)
- goal: only need O($n$) BGP connections, not O($n^2$)

# BGP communities

routes sent via BGP can have ‘communities’

extra information tagged on routes sent via BGP

large ISPs have lists of communities their customers/peers can use

… and these affect how those routes are used

# aside: Internet2

non-profit networking consortium

operates major US University-focused network

also makes eduroam work across different Universities

one of MARIA's major sources of connectivity (and indirectly one of UVA's major sources of connectivity)

has lots of peering relationships (incl. with big Internet companies)

(but not general Internet provider)

# selected Internet2 BGP communities

## From Internet2 Documentation:

### Internet2 External Traffic Influencing Communities

International, Non-International, or FEDNET peers may send the below community and we will set their localpref to 460 or 560 respectively:

Default - local-pref 500

11537:40 - Low (local-pref 460)

11537:160 - High (local-pref 560)

Connectors may send the below community and we will set their localpref to 540 or 620 respectively:

Default - local-pref 600

11537:140 - Low (local-pref 540)

11537:260 - High (local-pref 620)

Internet2 Peers may send the following communities:

11537:2002 - Block prefix to commercial R&E peers.

Internet2 International (ITN) peers may send the below communities for path prepending:

65001:65000 - prepend x1

65002:65000 - prepend x2

65003:65000 - prepend x3

### Internet2 Internal Traffic Influencing Communities

Internet2-connected Gigapops and connectors with heterogeneous participants (i.e. not all are universities) may need to use these communities to help them pass prefixes appropriately to their participants. For example, connectors can pass all prefixes on to their university participants, but should not pass commercial or US Fednet prefixes to their commercial or Fednet participants.

11537:888 - Block to external (Internet2 uses this community to mark prefixes that will not be reannounced to external peers).

11537:902 - non-I2 R&E sites sponsored for connection by members.

11537:910 - sponsored educational groups (Primary state networks).

11537:950 - Internet2 Participants.

11537:2000 - Commercial research - Lab participant.

11537:2001 - Special Commercial prefix used for V6 or multicast.

11537:2500 - Block advertisement to international.

11537:2501 - International.

....

### NET+ BGP Communities

Internet2 has implemented a set of BGP communities that, along with your own local BGP import policy, will allow you to control your traffic path to/from the Internet2 NET+ peer networks. All of these communities are specific to individual peer networks and do not affect routing to/from the other peer networks.

11537:5000 - Internet2 NET+ Services.

11537:5002 - Cloud.

....

The following community combination of <CODE>:<ASN> allows you to block or prepend prefixes sent to individual NET+ peers.

Codes:

- 65000 - prefixes will not be sent to NET+ peer's ASN
- 65001 - prefixes will be prepended 1 time to NET+ peer's ASN
- 65002 - prefixes will be prepended 2 times to NET+ peer's ASN
- 65003 - prefixes will be prepended 3 time2 to NET+ peer's ASN

NET+ peer ASN's

- 16509 - Amazon
- 62715 - Code42
- ...

# community options from prev slide

setting local-pref:

- you can decide how preferred your route is by Internet2
- maybe to make one primary, another secondary?

blocking route from being sent to specific place

prepending Internet2's AS before forwarding prefix

- hopefully make that route less preferred by others

prepending Internet2's AS before forwarding prefix to specific place

- hopefully make that route less preferred by that place

# other things with communities

Internet2 also uses communities to mark…

what location routes were learned from

what type of organization routes were learned from

whether Internet2 is only allowed to use the route non-commerically or not

…

# AS7007

## https://seclists.org/nanog/1997/Apr/444

### 7007 Explanation and Apology

**From**: “Vincent J. Bono” <vbono () MAI NET>
**Date**: Sat, 26 Apr 1997 19:41:35 EST

```
Dear All,

    I would like to sincerely apologize to everyone everwhere who
experienced problems yesterday due to the 7007 AS announcements.

    If anyone cares to know, here is what happened:

    At 11:30AM, EST, on 25 Apr 1997, our border router, stamped with
AS 7007, recieved a full routing view from a downstream ISP (well, a
view contacing 23,000 routes anyway).

    There was no distibute list imposed on the downstream since they
also advertise their customer AS's to us (they were also
experimenting with sending some routes out through us and some out
through the MAE).  We did filter out routes from them containing any
of our AS numbers but since they got the view from someone at
MAE-East none of our internal AS numbers showed up at all.  Not
having a filter imposed on the inbound side was our error.

    In an as yet unexplained twist of bits, the 7007 router then
began to de-aggregate the 23K route view *and* strip the AS path out
of it.  I will emphasize that we were running no IGP at the time.
Not one.  Not OSPF, not RIP, nothing.

...
```

# 2008 Pakistan Youtube

Pakistan Telecom recieved gov’t order to block youtube

implemented by inserting route for YouTube’s IP in internal network

misconfiguration meant route was advertised on BGP

was more specific than YouTube’s route, so made YouTube unreachablej

# timeline from RIPE NCC

https://www.ripe.net/about-us/news/youtube-hijacking-a-ripe-ncc-ris-case-study/

Youtube is announcing 208.65.152.0/22

18:47Z: Pakistan Telecom starts announcing 208.65.153.0/24

20:07Z: Youtube starts announcing 208.65.153.0/24

20:18Z: Youtube starts announcing 208.65.153.0/25 and 208.65.153.128/25

20:51Z: Pakistan Telecom's ISP forwards their announcements with additional copy of Pakistan Telecom's AS number

21:01Z: Pakistan Telecom's ISP withdraws routes initiated by Pakistan Telecom (but not Pakistan Telecom's customers)

# BGP Hijacking targeted cryptocurrency stuff

KLAYswap (Feb 2022), Celer Bridge (Sep 2022)

attackers intentionally redirected traffic to malicious version of services

… and stole money

both probably spoofed the final AS number in AS path

sometimes involved adding attacked IP range to routing registry

# nation-states?



# a more recent example



HACKING INTERNET ROUTING

## BGP hijack infecting networks caused by a comedy of errors that's not funny at all

What can we learn from a BGP hijacking that poisoned production software? Plenty.

DAN GOODIN – SEP 2, 2026 7:00 AM | 68



Credit: Getty Images

# route security

historically, no verification routes announced by “owner” of IP addresses

Internet Routing Registries — database of AS to IP address

- used automatically to filter out mistakes
- (not really designed to resist malicious attacks)

some verify which IPs they should have with RIRs

- “letter of agency/authority” to delegate

effort to deploy RPKI — public-key based scheme to verify routes

- checks that routes originated at correct AS
- doesn’t verify intermediate ASes will forward correctly

Internet Engineering Task Force (IETF)
Request for Comments: 6480
Category: Informational
ISSN: 2070-1721

M. Lepinski
S. Kent
BBN Technologies

February 2012

# An Infrastructure to Support Secure Internet Routing

## Abstract

This document describes an architecture for an infrastructure to support improved security of Internet routing. The foundation of this architecture is a Resource Public Key Infrastructure (RPKI) that represents the allocation hierarchy of IP address space and Autonomous System (AS) numbers; and a distributed repository system for storing and disseminating the data objects that comprise the RPKI, as well as other signed objects necessary for improved routing security. As an initial application of this architecture, the document describes how a legitimate holder of IP address space can explicitly and verifiably authorize one or more ASes to originate routes to that address space. Such verifiable authorizations could be used, for example, to more securely construct BGP route filters.

# partial tables

dealing with full Internet routing table is expensive

- 2026: around 1.3M prefixes (IPv4), 340k prefixes (IPv6)
- not quite as bad after “aggregation”
  - example: if 1.2.2/24 and 1.2.3/24 have same next hop, combine into 1.2.2/23

common shortcut if you have a couple ISPs:

keep ‘short’ routes (example: short AS path)

- to one of your “primary” ISPs
- maybe using ECMP

have default route for other cases

- special route for 0.0.0.0/0, ::/0
- to one of your ISPs

take advantage of more specific routes beating less specific

- typically also true in OSPF, RIP, etc.

# backup slides